

MaaHealth

Maternal Nutrition & Wellness Platform for Indian Mothers

Product Design Document

Research Findings → Features → Use Cases → Interaction Design

Team

Swarang Joshi - 2022114010
Devansh Singh - 2025204007
Dhawal P - 2025204005

1. Executive Summary

MaaHealth is a voice-first, regionally-aware maternal health and nutrition platform designed specifically for Indian mothers during pregnancy and the postpartum period. It bridges the critical gap between traditional Indian food culture and evidence-based nutritional guidance, making expert-backed dietary advice accessible in local languages, across geographies, and through familiar interaction modalities.

India accounts for roughly 17% of global maternal deaths, with malnutrition and poor dietary guidance identified as key contributing factors. Despite the availability of government schemes like POSHAN Abhiyaan, the last-mile gap in personalised, culturally-sensitive nutrition advice remains enormous. MaaHealth directly addresses this through a combination of AI-powered features, regional food databases, and human expert connectivity.

2. Research Context & Key Findings

Primary research conducted across semi-urban and urban Indian households through structured interviews, observational sessions, and surveys with pregnant women, new mothers, ANMs (Auxiliary Nurse Midwives), and ASHA workers surfaced the following critical insights:

2.1 Key User Insights

- **Insight 1 – Language Barrier:** Over 74% of mothers in Tier-2/3 cities and rural areas reported feeling confused by health app content written in English or formal Hindi. They abandoned apps after one or two sessions.
- **Insight 2 – Regional Diet Disconnect:** Generic apps recommend dal-roti or south Indian staples universally, ignoring Odia, Bengali, Rajasthani, or north-eastern cuisines. Mothers reported the suggestions were 'not our food.'
- **Insight 3 – Nutrition Knowledge Gaps:** Most mothers could not identify which trimester-specific nutrients they were deficient in or why. Iron, folate, and calcium requirements were poorly understood.
- **Insight 4 – Traditional vs. Medical Conflict:** Many mothers follow maayke (maternal home) dietary traditions that may conflict with clinical advice, leading to confusion and distrust of digital health tools.
- **Insight 5 – Access to Dieticians:** Only 11% of surveyed mothers had ever consulted a dietician. Cost, geographic distance, and lack of awareness were the primary barriers.
- **Insight 6 – Meal Sourcing Challenges:** Postpartum mothers noted difficulty in organising fresh, nutritious meals while managing a newborn. They relied on family or neighbours for home-cooked meals.
- **Insight 7 – Forgetfulness During Postpartum:** Postpartum mothers frequently missed medication and supplement doses due to sleep deprivation and cognitive load.
- **Insight 8 – Financial Constraints:** Weekly dietary planning budget was a key concern, especially in lower-middle-income households.
- **Insight 9 – Feedback Loop Absence:** Women expressed a desire for iterative improvement of their diet plans based on how they felt each week, but no existing tool facilitated this.

2.2 User Personas

Based on the research, three primary personas emerged:

Attribute	Priya (Pregnant, Urban)	Savitri (Postpartum, Semi-Urban)	Pravara (Postpartum, Urban)
Age	27	24	28
Language	Hindi / English	Telugu / Hindi	Marathi
Tech Literacy	Medium-High	Medium	Medium-High (Uses ChatGPT for medical translation and apps for soothing music)
Key Pain Point	Conflicting advice, confusing apps	Finding local nutritious meals post-delivery	High blood pressure complications, severe mood swings, and confusing medical jargon in sonography reports
Goal	Healthy pregnancy diet plan	Recover nutrition, breastfeed well	Safely manage health complications (BP), maintain mental stability, and understand medical reports in her native language

3. Product Features

Each feature below is directly motivated by one or more research insights. For each, the relevant interaction design / usability principle is also identified.

Feature F1: Voice-First Regional Language Interface

RESEARCH MOTIVATION

Insight 1 (Language Barrier) and Insight 2 (Regional Diet Disconnect). 74% of mothers in Tier-2/3 cities struggled with English-only apps. Voice input removes the literacy barrier entirely.

DESCRIPTION

The platform's primary interaction mode is voice. Users can speak in their regional language (support for Hindi, Telugu, Tamil, Bengali, Odia, Marathi, Kannada, Gujarati, and Punjabi) to query nutrition information, log meals, set reminders, and consult the symptom checker. The interface uses large tap targets, pictorial menus, and audio feedback to ensure accessibility for mothers with low digital literacy.

DESIGN PRINCIPLES APPLIED

- Universal Design / Accessibility: Voice-first + multilingual removes literacy, language, and motor barriers.
- Error Prevention: Confirmatory voice playback lets users verify before submitting.
- Recognition over Recall (Nielsen Heuristic 6): Visual-plus-audio prompts replace text-heavy menus.

Feature F2: Traditional & Science Bridge (Fun Facts)

RESEARCH MOTIVATION

Insight 4 (Traditional vs. Medical Conflict). Mothers trusted traditional dietary practices passed down by family but were confused when they conflicted with clinical advice.

DESCRIPTION

A contextual 'Did You Know?' section appears alongside every food recommendation. Each card shows: (a) the traditional belief or practice around that food, (b) the scientific explanation validating or gently correcting it, and (c) a fun fact about the nutrient. For example: 'Your dadi was right! Methi (fenugreek) seeds are rich in iron and phytoestrogens that support milk production.'

DESIGN PRINCIPLES APPLIED

- Cultural Sensitivity: Respects traditional knowledge while layering evidence-based corrections.
- Trust Building: Aligns with the mental model of the user's existing belief system before proposing change.
- Consistency & Standards: Uses a uniform card template so users learn the format quickly.

Feature F3: Regional Food & Nutrition Database

RESEARCH MOTIVATION

Insight 2 (Regional Diet Disconnect). Generic national databases miss regional staples like pittha (Jharkhand), litti-chokha (Bihar), or kozhukattai (Tamil Nadu).

DESCRIPTION

A curated, community-contributed database of 2,000+ Indian regional foods, tagged by state, season, availability, and nutritional profile. Mothers can search by ingredient or dish name in their regional language. Each entry shows macro and micronutrient values, trimester/postpartum suitability ratings, and substitution suggestions for unavailable items.

DESIGN PRINCIPLES APPLIED

- Match between System and Real World (Nielsen Heuristic 2): Uses food names as the user knows them, not clinical ingredient codes.
- Flexibility & Efficiency: Advanced filters for region, season, complication flags.
- Aesthetic & Minimalist Design: Only the most relevant nutritional fields are shown by default; deep detail is progressive disclosure.

Feature F4: Personalised Dietary Planning Engine

RESEARCH MOTIVATION

Insights 1, 2, 3, 5, 8. Mothers needed dietary plans that respect their region, complications, budget, and trimester.

DESCRIPTION

An onboarding flow collects: (a) current week of pregnancy or days post-delivery, (b) region/state and preferred cuisine, (c) dietary restrictions (vegetarian, Jain, allergies), (d) medical complications (gestational diabetes, anaemia, thyroid conditions), and (e) weekly food budget. The engine generates a 7-day meal plan with breakfast, lunch, snacks, and dinner, sourced from the Regional Food Database, and adjusts caloric and micronutrient targets to clinical guidelines (ICMR-NIN recommendations).

DESIGN PRINCIPLES APPLIED

- User Control & Freedom: Users can swap individual meals, mark dislikes, or regenerate a full day.
- Visibility of System Status: A weekly nutrition-coverage progress bar shows which nutrients are on track.
- Error Recovery: If the user removes a meal, the system suggests an equivalent alternative.

Feature F5: AI Meal Photo Analyser

RESEARCH MOTIVATION

Insight 3 (Nutrition Knowledge Gaps). Women wanted a quick, effortless way to know if what they just ate was nutritionally adequate.

DESCRIPTION

Mothers take a photo of their plate. AI identifies the visible foods, estimates portion sizes using visual cues, and provides an instant rough nutritional breakdown. The output shows: calories, protein, iron, calcium, folate, and fibre. A colour-coded indicator highlights nutrients that are below the daily target. A 'What to add tomorrow' prompt suggests one or two easily available foods to address the gap.

DESIGN PRINCIPLES APPLIED

- Immediate Feedback (Nielsen Heuristic 1): Analysis is returned within seconds; progress animation reassures the user.
- Closure: The 'what to add tomorrow' prompt closes the cognitive loop and converts insight into action.

- Direct Manipulation: The photo-as-input modality is the most natural way for non-literate users to log food.

Feature F6: Doctor-Validated AI Symptom Checker

RESEARCH MOTIVATION

Insight 3 (Nutrition Knowledge Gaps) and Insight 5 (Access to Dietitians). Mothers often Google symptoms, encountering unreliable or alarming information.

DESCRIPTION

A conversational symptom intake screen (voice or text) asks the user to describe what she is feeling. A validated decision-tree model, reviewed and updated by a panel of OBGYNs and nutritionists, returns a triage output: (a) likely nutritional cause (e.g., fatigue linked to iron deficiency), (b) dietary recommendation, and (c) a flag if the symptom requires in-person medical attention. All outputs include a disclaimer and physician sign-off watermark.

DESIGN PRINCIPLES APPLIED

- Safety by Design: The system defaults to 'consult a doctor' for any ambiguous symptom.
- Help & Documentation: Expandable explanations for why a nutrient might cause a symptom.
- Trust Signifiers: Display of validating doctor profiles increases credibility.

Feature F7: Dietician & Doctor Connect

RESEARCH MOTIVATION

Insight 5 (Access to Dietitians). Only 11% of mothers had ever consulted a dietician, primarily due to cost and geography.

DESCRIPTION

An in-app marketplace lists empanelled dietitians and nutritionists, filterable by language, specialisation (gestational diabetes, PCOS, anaemia), and consultation fee. Mothers can book 15-minute video or voice consultations. Integration with the personalised plan means the dietician sees the user's meal history, deficiency flags, and complication profile before the call. Post-consultation, the dietician can push a revised weekly plan directly into the app.

DESIGN PRINCIPLES APPLIED

- End-to-End Task Support: The feature handles discovery, booking, context-sharing, and plan update in one flow.
- Learnability: The first consultation is subsidised with a government or NGO voucher to lower the entry barrier.
- Trust & Social Proof: Dietician profiles include patient reviews, languages spoken, and FSSAI/NIN credentials.

Feature F8: Home Cook Connect via WhatsApp

RESEARCH MOTIVATION

Insight 6 (Meal Sourcing Challenges). Postpartum mothers struggle to cook nutritionally adequate meals themselves.

DESCRIPTION

A hyperlocal network of verified home cooks (primarily women in the same locality) is accessible via in-app listing and WhatsApp deep-link. Mothers browse cook profiles, view sample menus, and initiate a WhatsApp conversation to place meal prep orders. The cook receives a recipe card auto-generated from the mother's current weekly plan, showing exact portions and ingredient notes. Order status and cost are tracked in-app.

DESIGN PRINCIPLES APPLIED

- Familiar Interaction Modality: WhatsApp is India's de facto communication channel, reducing the learning curve to zero.
- Community & Social Support: Reinforces neighbourhood trust networks, aligning with cultural norms.
- Progressive Disclosure: In-app summary; full conversation happens in WhatsApp.

Feature F9: Smart Reminder System

RESEARCH MOTIVATION

Insight 7 (Forgetfulness During Postpartum). Sleep-deprived mothers frequently missed supplements and medications.

DESCRIPTION

A configurable reminder system sends push notifications and voice alerts (in the user's language) for: (a) daily supplements and medications with dose details, (b) meal times aligned to the weekly plan, and (c) daily water intake nudges. Reminders adapt to the time zone and detected sleep patterns (via opt-in screen-lock data). Snooze options are large tap targets to minimise friction when a mother is holding a baby.

DESIGN PRINCIPLES APPLIED

- Error Prevention: Timely reminders pre-empt missed doses rather than correcting them after.
- Flexibility: Users can set custom reminder schedules; bulk-edit all times in one screen.
- Minimalist Interaction: One-tap 'Mark as done' on the notification surface.

Feature F10: Weekly Feedback & Plan Iteration Loop

RESEARCH MOTIVATION

Insight 9 (Feedback Loop Absence). Women wanted their diet plans to evolve based on how they felt each week.

DESCRIPTION

Every Sunday, a brief 4-question check-in (voice or tap) asks: (a) energy levels this week, (b) any new symptoms, (c) foods the user enjoyed/disliked, and (d) budget adherence. Responses feed back into the personalisation engine to adjust next week's plan. The in-app timeline shows how the plan has evolved over weeks, reinforcing a sense of progress. A monthly summary is shareable with the dietician.

DESIGN PRINCIPLES APPLIED

- Iterative Personalisation: System becomes more accurate over time without requiring manual re-profiling.
- User Engagement & Motivation: Progress timeline creates a positive feedback loop.
- Short Interaction Bursts: Check-in designed for under 2 minutes to respect a mother's cognitive and time load.

4. Feature–Research Insight Mapping

The table below maps each feature to its primary research insight, enabling traceability from user need to product decision.

Feature	Primary Insight(s)	Core User Need Addressed	Interaction Design Principle
F1 – Voice Interface	Insight 1, 2	Language & literacy accessibility	Universal Design, Recognition over Recall
F2 – Science Bridge	Insight 4	Trust between tradition and medicine	Mental Model Alignment, Cultural Sensitivity
F3 – Regional DB	Insight 2	Regionally relevant food information	Match with Real World, Localisation
F4 – Diet Planning	Insight 1,2,3,5,8	Personalised, affordable weekly meal plan	User Control, Visibility of System Status
F5 – Photo Analyser	Insight 3	Effortless meal nutritional feedback	Immediate Feedback, Direct Manipulation
F6 – Symptom Checker	Insight 3, 5	Reliable health guidance without a doctor visit	Safety by Design, Trust Signifiers
F7 – Dietician Connect	Insight 5	Affordable access to expert advice	End-to-End Task Support, Social Proof
F8 – Cook Connect	Insight 6	Outsourcing meal prep locally	Familiar Modality (WhatsApp), Community
F9 – Reminders	Insight 7	Medication and nutrition adherence	Error Prevention, Minimal Friction
F10 – Feedback Loop	Insight 9	Plan evolution over time	Iterative Personalisation, Motivation

5. Use Cases

Each use case follows the standard structure: User Goal, Pre-conditions, Main Interaction Flow, Alternative/Exception Flows, and Expected Outcome.

UC-01: First-Time Onboarding in Regional Language

Actor	Savitri, 24, Telugu-speaking, 3 days post-delivery
Goal	Set up her MaaHealth profile and receive a personalised postpartum meal plan in Telugu.
Feature	F1 (Voice Interface), F3 (Regional DB), F4 (Diet Planning)
Pre-condition	Savitri has downloaded the app. She selects Telugu on the splash screen.
Main Flow	1. App greets her in Telugu with a voice prompt: 'Meeru delivery ayyara? Pregnancy lo unnara?' (Did you just deliver? Are you pregnant?)2. Savitri taps 'Just delivered'. App asks her delivery date and her home district (Guntur).3. Voice prompt asks about dietary restrictions. Savitri says 'I am vegetarian'.4. App asks about any complications (gestational diabetes, anaemia). Savitri taps 'Anaemia'.5. App asks approximate weekly food budget (slider with rupee values). She sets Rs. 800.6. App generates a 7-day postpartum meal plan using Andhra staples: pesarattu, pappu charu, nuvvula laddu, methi rice.7. The plan is read aloud in Telugu; Savitri can swipe through daily cards.
Exception Flow	If Savitri does not know her delivery date, she can say 'about 3 days back' and the app estimates.
Outcome	Savitri has a culturally relevant, medically appropriate 7-day meal plan in her language within 4 minutes.
Usability Principles	Recognition over Recall (voice prompts, not typed forms); User Control (can redo any step); Error Tolerance (approximate date input accepted).

UC-02: Meal Photo Scan During Lunch

Actor	Priya, 27, pregnant (28 weeks), urban, Hindi-speaking
Goal	Quickly check if her lunch of palak paneer, roti, and salad meets her pregnancy nutritional needs.
Feature	F5 (AI Meal Photo Analyser), F4 (Diet Planning), F3 (Regional DB)
Pre-condition	Priya is logged in and her 28-week pregnancy profile is active.
Main Flow	1. Priya taps the camera icon on the home screen.2. She photographs her plate of palak paneer (150g estimated), 2 rotis, and a kachumber salad.3. AI returns analysis in 4 seconds: Calories: 520 kcal, Protein: 18g, Iron: 4.2mg, Calcium: 210mg, Folate: 90mcg.4. A colour-coded bar shows Iron (yellow: 42% of daily target), Folate (red: 23% of target), Calcium (green: 70%).5. Prompt: 'Add a small bowl of chana dal at dinner for folate, and a glass of milk for calcium'.6. Priya taps 'Add to today's log'. The log updates her daily tracker.
Exception Flow	If the AI is uncertain (e.g., a mixed dish like pulao), it shows the top 3 possible identifications and asks the user to confirm.
Outcome	Priya knows her nutritional gaps in under 30 seconds and has an actionable dinner suggestion.
Usability Principles	Immediate Feedback; Visibility of System Status (colour-coded progress bars); Actionable Closure (tomorrow suggestion).

UC-03: Consulting the Traditional & Science Bridge

Actor	Meena, 22, Odia-speaking, 2 weeks postpartum, rural
Goal	Understand whether she should follow her mother-in-law's advice to avoid eating fish for 40 days postpartum.
Feature	F2 (Science Bridge), F1 (Voice Interface)
Pre-condition	Meena is on the food detail screen for 'rohu fish'.
Main Flow	1. Meena asks via voice: 'Macha khaba jochi ki nahin?' (Is it okay to eat fish?)2. The Science Bridge card appears, titled 'Ghare katha vs. Vigyan' (Home wisdom vs. Science).3. Traditional belief shown: 'Many Odia families advise avoiding fish for 40 days postpartum.'4. Science explanation: 'Rohu fish is rich in DHA and omega-3 fatty acids that support your baby's brain development through breast milk. The 40-day restriction has no clinical basis. Small portions of fresh, well-cooked fish 2-3 times per week are beneficial.'5. Fun fact: 'DHA in mother's milk can improve infant visual acuity in the first year.'6. A 'Share with family' button lets Meena send the card to her mother-in-law via WhatsApp.
Exception Flow	If Meena has a documented fish allergy in her profile, the card recommends alternative omega-3 sources like flaxseed and walnuts instead.
Outcome	Meena gains confidence to make an informed dietary decision and can diplomatically share scientific backing with her family.
Usability Principles	Mental Model Alignment; Cultural Sensitivity; Shareability (social proof for family persuasion).

UC-04: Booking a Dietician Consultation

Actor	Priya, 27, pregnant (32 weeks), diagnosed with gestational diabetes
Goal	Consult a Hindi-speaking dietician who specialises in gestational diabetes.
Feature	F7 (Dietician Connect), F4 (Diet Planning), F10 (Feedback Loop)
Pre-condition	Priya's profile includes a gestational diabetes flag. She has 3 weeks of meal log history.
Main Flow	1. Priya taps 'Consult Expert' on her nutrition dashboard.2. Filter pre-set to 'Gestational Diabetes + Hindi'. Three dieticians are shown with ratings, fees (Rs. 200-500), and next available slots.3. Priya selects Dr. Nisha Sharma (Rs. 300, rated 4.8, next slot in 2 hours).4. App shows Dr. Nisha: 'Your last 3 weeks of meal logs and your blood sugar trends will be shared with Dr. Nisha before the call.'5. Priya confirms and pays via UPI.6. At the scheduled time, a video call launches in-app. Dr. Nisha has reviewed the meal history.7. Post-call, Dr. Nisha pushes a revised low-GI meal plan for week 33 directly into the app.8. Priya receives a push notification: 'Your Week 33 plan has been updated by Dr. Nisha.'
Exception Flow	If no dietician is available within 4 hours, the app offers an asynchronous 'plan review' option where the dietician reviews and responds within 12 hours.
Outcome	Priya receives a clinically revised plan specific to her gestational diabetes, informed by her real meal data.
Usability Principles	End-to-End Task Flow; Context Continuity (meal history pre-shared); Trust Signals (credentials, reviews).

UC-05: Ordering a Meal from a Home Cook via WhatsApp

Actor	Savitri, 24, 2 weeks postpartum, Guntur
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Goal	Order today's postpartum lunch (methi khichdi, rasam, curd) from a nearby home cook.
Feature	F8 (Cook Connect), F4 (Diet Planning)
Pre-condition	Savitri's Week 2 postpartum plan shows today's lunch as methi khichdi + rasam + curd.
Main Flow	1. On the meal card for today's lunch, Savitri taps 'Find a Cook nearby.'2. Map shows 3 verified home cooks within 2 km. Each profile shows rating, languages spoken, and sample menu photo.3. Savitri taps 'Order via WhatsApp' on Lakshmi Amma (rated 4.7, Telugu-speaking, 0.8 km away).4. WhatsApp opens with a pre-filled message in Telugu, including today's recipe card (dish name, portion size, key ingredients from her plan).5. Lakshmi Amma confirms availability and price (Rs. 80).6. Savitri marks the order as placed in-app; a delivery time reminder is set.7. Post-meal, Savitri taps a one-tap rating for Lakshmi Amma in-app.
Exception Flow	If no cook is available within 5 km, the app shows the recipe and suggests a simplified 20-minute self-cook version with step-by-step voice instructions.
Outcome	Savitri receives a nutritionally aligned, home-cooked postpartum meal without having to cook herself, using a familiar WhatsApp workflow.
Usability Principles	Familiar Interaction Modality; Reduced Cognitive Load; Community Trust Networks.

UC-06: Weekly Feedback Check-In

Actor	Priya, 27, 34 weeks pregnant
Goal	Provide weekly feedback so her plan adjusts to her changing needs and preferences.
Feature	F10 (Feedback Loop), F4 (Diet Planning), F9 (Reminders)
Pre-condition	It is Sunday. Priya has completed Week 4 of using MaaHealth. A push notification prompts her.
Main Flow	1. Sunday 10am: push notification 'How was your week, Priya? 2-minute check-in.'2. Screen 1: 'How were your energy levels this week?' (5-tap emoji scale, very low to very high).3. Screen 2: 'Any new symptoms?' (pre-listed chips: nausea, swelling, heartburn, or type your own).4. Screen 3: 'Which meals did you enjoy most / least?' (thumbs up/down on the week's top 5 meals).5. Screen 4: 'Did you stay within your weekly budget?' (Yes / Almost / No).6. App processes responses: Priya reported low energy + heartburn, disliked rajma.7. Week 35 plan replaces rajma with moong dal (easier to digest), adds iron-rich foods for low energy, and reduces spicy items for heartburn.
Exception Flow	If Priya skips the check-in, a gentle reminder is sent Monday morning. After 3 missed weeks, the app nudges her with 'Your plan hasn't been updated in a while.'
Outcome	Priya's Week 35 plan is automatically updated to address her reported symptoms and food preferences without needing a dietician consultation.
Usability Principles	Iterative Personalisation; Short Interaction Bursts (<2 min); Positive Reinforcement (progress timeline shown after completion).

UC-07: AI Symptom Checker for Fatigue

Actor	Meena, 22, 3 weeks postpartum, Odia-speaking
Goal	Understand why she has been feeling constantly tired and dizzy.
Feature	F6 (Symptom Checker), F1 (Voice Interface)

Pre-condition	Meena is on the home screen and taps the 'Health' tab.
Main Flow	1. Meena taps 'Feeling unwell?' and selects voice input.2. She says 'Mu bahut thakila laguchi, matha ghuruchi' (I feel very tired, getting dizzy).3. App recognises the Odia input and displays a follow-up: 'How many days? Are you eating iron-rich foods?'4. Meena says 'About a week. I mostly eat rice and dal.'5. The symptom checker flags: 'These symptoms are consistent with postpartum iron-deficiency anaemia, which is common in new mothers.'6. Dietary recommendation displayed: add drumstick leaves (sahajan), beetroot, and jaggery to daily diet. Avoid tea/coffee immediately after meals as it inhibits iron absorption.7. Alert: 'If dizziness is severe or you feel breathless, please consult a doctor immediately. Tap here to find a nearby clinic.'8. Option to share the symptom summary with her ASHA worker.
Exception Flow	If Meena mentions chest pain or severe headache, the system bypasses dietary advice and immediately displays emergency contact numbers.
Outcome	Meena receives a likely nutritional explanation for her symptoms, an actionable dietary change, and a clear safety escalation path.
Usability Principles	Safety by Design; Help & Documentation; Clear Escalation Path; Trusted Authority Signals (doctor-validated badge).

6. Interaction Design Framework

6.1 Heuristics Applied (Nielsen's 10 Usability Heuristics)

Nielsen Heuristic	Application in MaaHealth
1. Visibility of System Status	Weekly nutrition progress bars, meal log completion indicators, plan update notifications.
2. Match System & Real World	Food names in regional language, currency in INR, local measurement units (katori, glass).
3. User Control & Freedom	Swap any meal, cancel a reminder, undo meal log, re-do onboarding.
4. Consistency & Standards	Uniform card design for all food items, Science Bridge cards, and use case flows.
5. Error Prevention	Symptom checker safety flags; reminders pre-empt missed doses; photo analysis asks for confirmation.
6. Recognition over Recall	Voice prompts + visual icons; food choices shown as images, not text lists.
7. Flexibility & Efficiency	Voice shortcuts for power users; tap-flow for low-literacy users.
8. Aesthetic & Minimalist Design	Progressive disclosure: basic info first, nutritional depth behind a 'show more' tap.
9. Help Users Recognise & Recover Errors	Unclear food identified: top 3 options displayed. Budget exceeded: cheaper substitutes shown.
10. Help & Documentation	In-line 'Why?' buttons on every recommendation; ASHA worker contact card pinned in profile.

6.2 Mental Model & Trust Design

Indian mothers' mental models around food are deeply shaped by family traditions, Ayurvedic principles, and regional practices. MaaHealth's design respects this by:

- Never leading with clinical jargon. Every recommendation begins with the familiar food name ('your methi laddu'), not the nutrient.
- Validating before correcting. The Science Bridge always acknowledges the traditional belief before presenting the scientific position.
- Using trusted authority figures. Symptom checker outputs carry an OBGYN validation badge; dietician profiles show NIN certification.
- Social proof integration. Home cook reviews, dietician ratings, and community-shared Science Bridge cards leverage peer trust networks.

6.3 Accessibility & Inclusive Design

Given the diversity of users across literacy levels, languages, and digital familiarity, MaaHealth applies inclusive design throughout:

- Voice-first interaction reduces dependence on text literacy.
- Minimum touch target size of 44x44pt for all interactive elements (WCAG 2.1 AA).
- High-contrast colour palette tested against the most common environmental conditions (bright outdoor light common in rural settings).

- Offline-first architecture: core meal plan and reminder features work without internet; database syncs when connectivity is available.
- Low-data mode: images are compressed; audio responses are offered as an alternative to video.

6.4 Behaviour Change Design

MaaHealth incorporates behaviour change frameworks (BJ Fogg's Behaviour Model and Self-Determination Theory) to sustain engagement:

- Motivation: Weekly progress visualisation, milestone cards ('You completed 3 weeks of your iron plan'), and shareable achievements.
- Ability: Reducing friction at every step (one-tap meal log, pre-filled WhatsApp orders, voice input).
- Prompt: Timely, contextual nudges (reminder before usual meal time, not at a fixed daily time).

7. Conclusion

MaaHealth translates 10 primary research insights from Indian pregnant and postpartum mothers into a cohesive set of 10 interconnected features, seven of which are illustrated through detailed use cases covering first-time onboarding, meal scanning, traditional-science bridge consultation, dietician booking, home-cook ordering, weekly feedback, and symptom checking.

The design consistently applies Nielsen's usability heuristics, inclusive design principles, and India-specific cultural considerations to ensure that the platform serves Priya in urban Delhi, Savitri in semi-urban Guntur, and Meena in rural Odisha with equal effectiveness. By anchoring every product decision in a specific research finding, MaaHealth achieves traceability from user pain point to interaction design outcome, a standard that validates its readiness for prototype development and user testing.

Next steps include: (1) paper prototype testing with 10 mothers across 3 regional profiles, (2) A/B testing of voice versus tap-first onboarding flows, and (3) pilot deployment of the Regional Food Database with ASHA workers in 2 districts.

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